

Experiment (1)

FINE MEASUREMENTS

1.1 Introduction

Physics is an experimental science. Physics theories can provide predictions that can be verified through experiments. Experiments require measurements and measuring tools. Although in Physics problems we typically can calculate an exact solution, that is not the case for experimental measurements. The accuracy and precision of measurements are therefore important. If the measurement disagrees with the prediction, we must determine whether the source of discrepancy is an error in the experiment or an error in the theory under test.

Thus, mastering measuring tools and understanding the errors they introduce is vital for Physics and any Science. Moreover, these tools are indispensable for most professions, from technicians to experimentalists and from home workshops to factories and labs.

1.2 Objectives

- The student will be able to use measuring tools such as verniers, micrometers, balances, measuring tapes and stopwatches.
- The student will be able to distinguish between systematic and random errors.
- The student will be able to determine the zero errors of the measuring tools.
- The student will analyze errors in two examples. The first is to calculate the density of a cylinder by measuring its volume and mass. The second is to calculate the periodic time of a simple pendulum.

1.3 Theory

Measurement error is defined as the difference between the actual value of a quantity and the value obtained in measurement. The main types of errors are Systematic error and Random error.



In most real life situations, we do not have an independent means to know the “actual” value of a quantity. An important consideration in error analysis is how to estimate how far away we are from a quantity that we don’t know!

1.3.1 The density of a cylinder

The density ρ of any material is defined as the mass per unit volume of this material,

$$\rho = \frac{m}{V} \quad (1.1)$$

where m is the mass and V is the volume. The volume of the cylinder is calculated by:

$$V = \pi D^2 h / 4 \quad (1.2)$$

where D is the diameter of the cylinder and h is the height of the cylinder. If we have an estimate for the errors associated with measurements of m , D , and h , we can estimate the error for ρ using the procedures described in the Error Analysis chapter. In this case, the uncertainty formula is given by (see the appendix):

$$\left| \frac{\sigma_\rho}{\bar{\rho}} \right| = \sqrt{\left(\frac{\sigma_m}{\bar{m}} \right)^2 + \left(\frac{\sigma_h}{\bar{h}} \right)^2 + \left(2 \frac{\sigma_D}{\bar{D}} \right)^2} \quad (1.3)$$

where $\bar{m}$, $\bar{D}$ and $\bar{h}$ are the average values, and

$$\bar{\rho} = \frac{4 \bar{m}}{\pi \bar{D}^2 \bar{h}} \quad (1.4)$$

is the average value of the measured density (if the density has been calculated in this way from measured values for m , D , and h).

σ refers to the standard deviation which is given by,

$$\sigma = \sqrt{S} = \sqrt{\frac{\sum_i (\Delta x_i)^2}{N}} \quad (1.5)$$

where N is the number of trials.

1.4 Setup

The tools and materials required for Tasks (1), (2) and (3) are shown in Figure 1.1, Figure 1.2 and Figure 1.3 respectively.

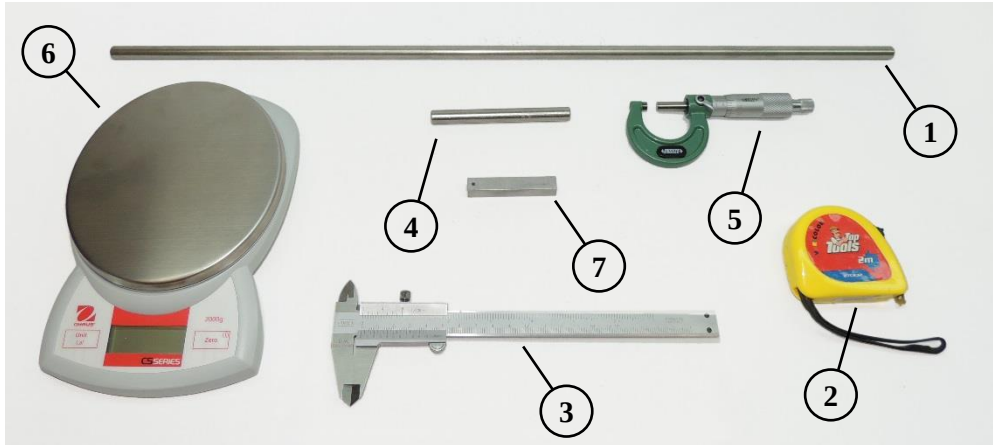


Figure 1.1: Task (1) tools and materials.



Figure 1.2: Task (2) tools and materials.

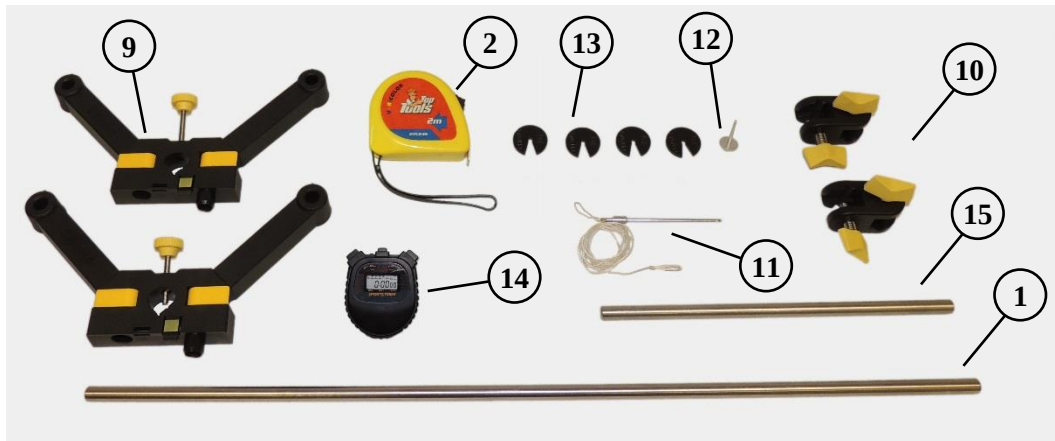


Figure 1.3: Task (3) tools and materials.

- | | |
|-----------------------|----------------------------|
| 1 Long rod | 9 Support base |
| 2 Measuring tape | 10 Boss head |
| 3 Vernier caliper | 11 Holding pin with string |
| 4 Short rod with hole | 12 Weights holder |
| 5 Micrometer | 13 Slotted masses |
| 6 Digital balance | 14 Stopwatch |
| 7 Aluminum column | 15 Short support rod |
| 8 Metallic cylinder | |

1.5 Procedure

In this section, we describe how to measure quantities such as m , D , h , as we defined them in the previous sections. From these measurements we will use our mathematical models to derive measurement for volume, density, etc. Then we will see how to estimate the error in these measurements. See the [Appendix](#) to learn about the measuring instruments and how to find the zero error for each of them.



On measuring any physical quantity, a number of trials should be done to reduce the error in the measurement as much as possible. The number of trials depends on the estimated or expected uncertainty in the measured value so you always need to use your judgement on how many trials to do. In the following tasks, it is recommended to have a minimum of $N=6$ trials for each measurement in order to practice how the number of trials affect the final measured value.

1.5.1 Task (1): Using the measuring instruments.

- 1- For the long rod (1), use the measuring tape (2) to measure its length as shown in Figure 1.4. Repeat this measurement several times for error analysis.



Figure 1.4: Measuring the length of the long rod using the measuring tape.



Although the vernier caliper (3) is of higher precision than the measuring tape, the length of the rod exceeds the maximum length that can be measured using the caliper. Therefore, it cannot be used for this measurement.

- 2- For the short rod with hole (4), use the measuring tape to measure its length. Repeat this measurement several times for error analysis.
- 3- Use the vernier caliper to measure the hole depth of this short rod (4) using the designated “stem” for measuring depths as shown in Figure 1.5. Repeat this measurement several times for error analysis.
- 4- Use the vernier caliper’s jaws dedicated for measuring the inner dimensions to measure the inner diameter of the short rod hole as shown in Figure 1.6. Repeat this measurement several times for error analysis.



Figure 1.5: Measuring the depth of the short rod hole using the vernier caliper.



Figure 1.6: Measuring the inner diameter of the short rod hole using the vernier caliper.

- 5- Place the rod between the vernier caliper's jaws dedicated for measuring the outer dimensions and measure the length of the short rod using the vernier caliper. Repeat this measurement several times for error analysis.



Figure 1.7 shows how to place the rod between the vernier caliper jaws to measure its length correctly.

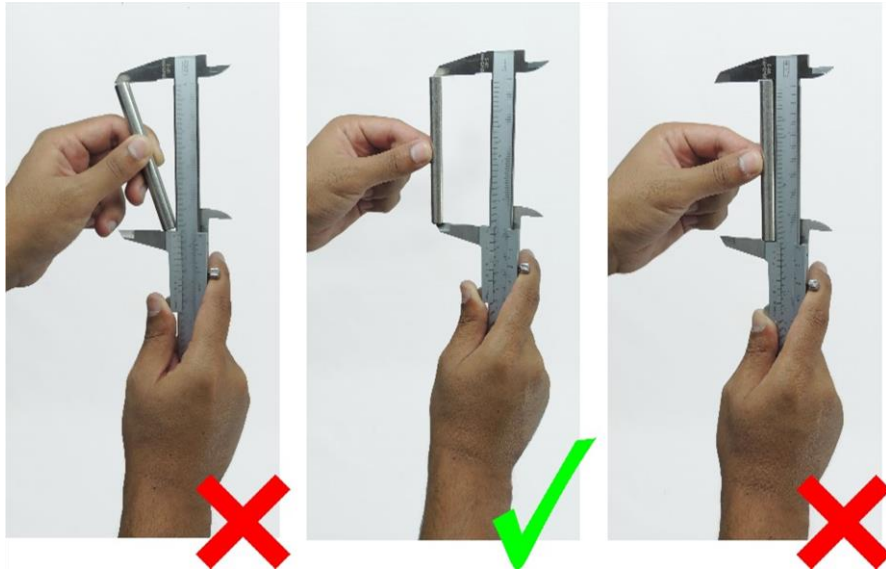
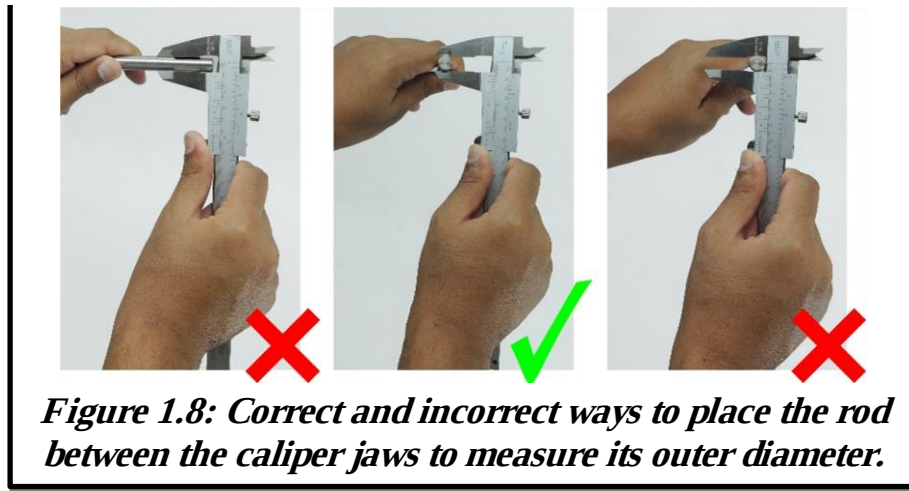


Figure 1.7: Correct and incorrect ways to place the rod between the caliper measuring jaws to measure its length.

- 6- Place this short rod between the vernier caliper's jaws dedicated for measuring the outer dimensions and measure its outer diameter. Repeat this measurement several times for error analysis.



Figure 1.8 show how to place the rod between the jaws of the vernier caliper to measure its outer diameter correctly.



- 7- Use the micrometer (5) to measure the outer diameter of the short rod. Figure 1.9 shows how to place the rod between the anvil and the spindle of the micrometer to get the best measurement. Repeat this measurement several times for error analysis.



Figure 1.9: Measuring the outer diameter of the rod using the micrometer

- 8- Use the digital balance (6) to measure the mass of this short rod and the Aluminum column (7). Repeat the measurement of each part several times for error analysis.
- 9- For the Aluminum column, use the measuring tape, vernier caliper and the micrometer to measure its height, length and width (if possible), as you did for the short rod with hole in steps 2 to 7. Repeat each measurement using each measuring tool several times for error analysis.

1.5.2 Task (2): Determination of the error in measuring the volume of a metallic cylinder.

- 1- Measure the zero errors in the vernier caliper (3), the micrometer (5), and the digital balance (6) (h_{Zero} , D_{Zero} , M_{Zero} , respectively). Check the [Appendix](#) for more details.
 - Use the mettalic cylinder (8) (Completely solid with no holes) to make the following measurements:
- 2- Measure the diameter of the cylinder using the micrometer several times for error analysis. Determine the average value $\bar{D}$ of these readings, then correct the zero error of the micrometer (if exists) then calculate the average value $\bar{D}$ after correction of the zero error (Check the [Appendix](#) for more details on zero errors).
- 3- Measure the height of the cylinder using the vernier caliper several times for error analysis. Determine the average value $\bar{h}$ of these readings, then correct the zero error of the vernier (if exists) then calculate the average value $\bar{h}$ after correction of the zero error.
- 4- Measure the mass of the cylinder using the digital balance several times for error analysis. Determine the average value $\bar{m}$ of these readings, then correct the zero error of the balance (if exists) then calculate the average value $\bar{m}$ after correction of the zero error.
- 5- Determine the average value of the density of the material of the cylinder using relation (1.4).
- 6- Determine the difference between the individual measurement of the diameter and the average value $\Delta D = |\bar{D} - D|$.
- 7- Repeat the previous two steps to get Δh and Δm .
- 8- Use relations (1.5) and (1.3) to determine the the uncertainty in measuring the density σ_ρ .

$$\left| \frac{\sigma_\rho}{\bar{\rho}} \right| = \sqrt{\left(\frac{\sigma_m}{\bar{m}} \right)^2 + \left(\frac{\sigma_h}{\bar{h}} \right)^2 + \left(2 \frac{\sigma_D}{\bar{D}} \right)^2}$$

1.5.3 Task (3): Measuring the periodic time of a simple pendulum.

- 1- Use the parts shown in Figure 1.3 to construct the simple pendulum shown in Figure 1.10. Try to adjust the height of the lower bosshead (14) and use the measuring tape (2) to make the total length of the pendulum from the holding pin to the center of mass of the weights (13) as close to 100 cm as possible.
- 2- Use the measuring tape again to measure the length l of the pendulum from the holding pin to the center of mass of the weights adjusted previously as accurately as possible.

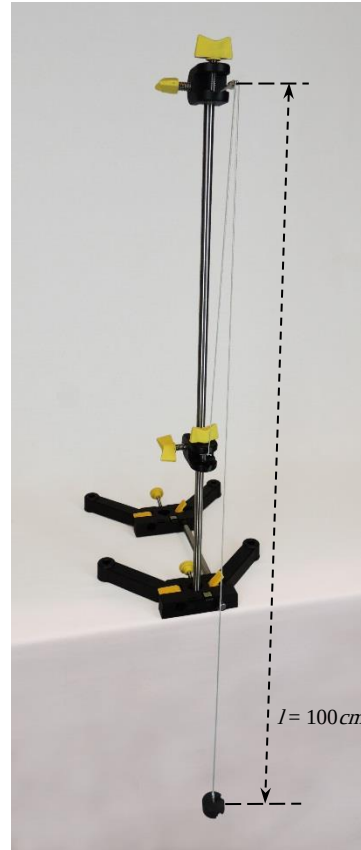


Figure 1.10: Simple pendulum.



Although the length from the holding pin to the center of mass of the weights was adjusted to be as close to 100 cm as possible, there is no doubt that an error was introduced in this adjustment. To reduce this error, multiple measurements of the pendulum length should be taken.

- 3- Reset the stopwatch (14) to have “zero” (00:00:0000) reading.

- 4- Move and hold the end of the pendulum about 10 cm laterally as shown in Figure 1.11.
- 5- Release the weight holder carefully to allow the pendulum to oscillate parallel to the edge of the table (repeat this step if it is not oscillating parallel to the table), at the same instant, start the stopwatch (14) by pressing the “start/stop” button.



Figure 1.11: Move the end of the pendulum about 10 cm laterally .



To obtain the best results:

- When moving the end of the pendulum laterally, hold it from the weight holder (12) not from the string as shown in Figure 1.12.

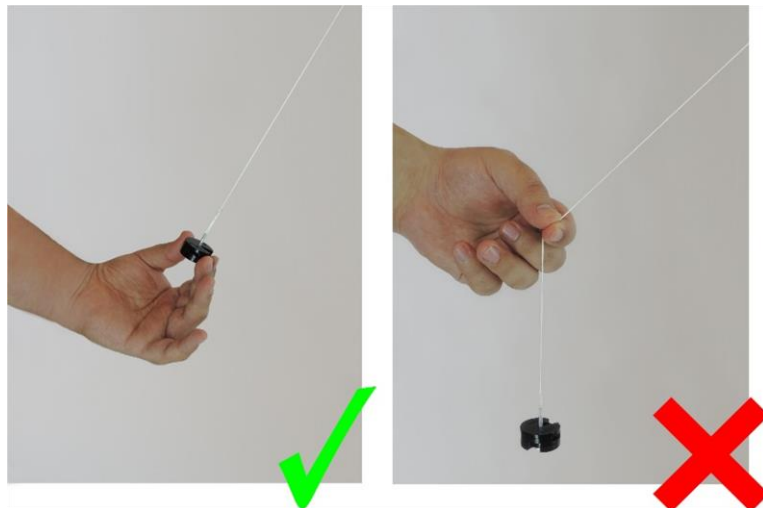


Figure 1.12: Hold the pendulum from the weight holder not from the string.

- When the pendulum is moved, the string, the weight holder and the tip of the pin (11) should all be found in the same vertical plane parallel to the table front side as shown in Figure 1.13.

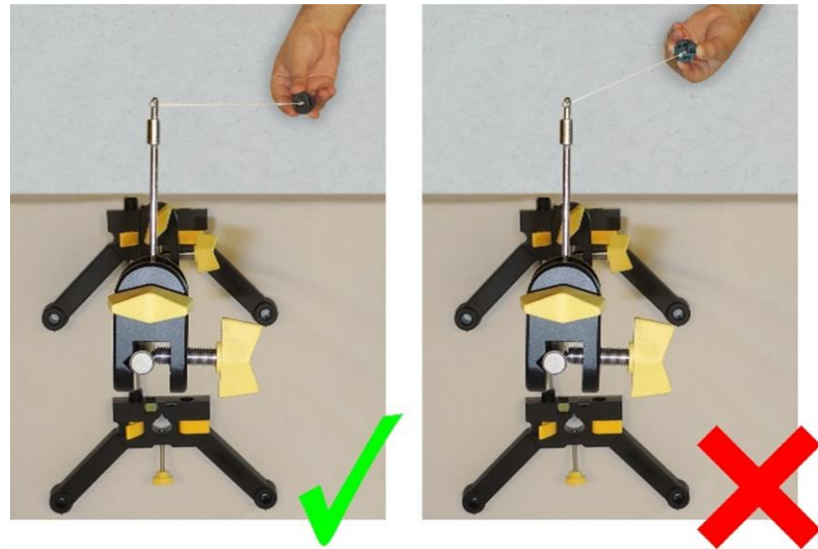


Figure 1.13: The string, the weight holder and the tip of the pin should all be found in the same vertical plane .

- 6- When the pendulum completes 20 entire oscillations, press the stopwatch “start/stop” button again. Read the time from the watch. This is the time taken to complete 20 oscillations t_{20} . Then calculate the time of one oscillation $T = t_{20} / 20$.
- 7- Repeat steps 3 to 6 several times for error analysis.
- 8- Determine the average value of the periodic time. Then correct the zero error of the stopwatch (if exists) and calculate the corrected average value $\bar{T}$ and the standard deviation σ_T .

Appendix 1

Using measuring instruments

1- Vernier caliper

A vernier caliper as the one shown in Figure 1.14 has a vernier scale which allows a length to be read to $1 / 10 \text{ mm}$.

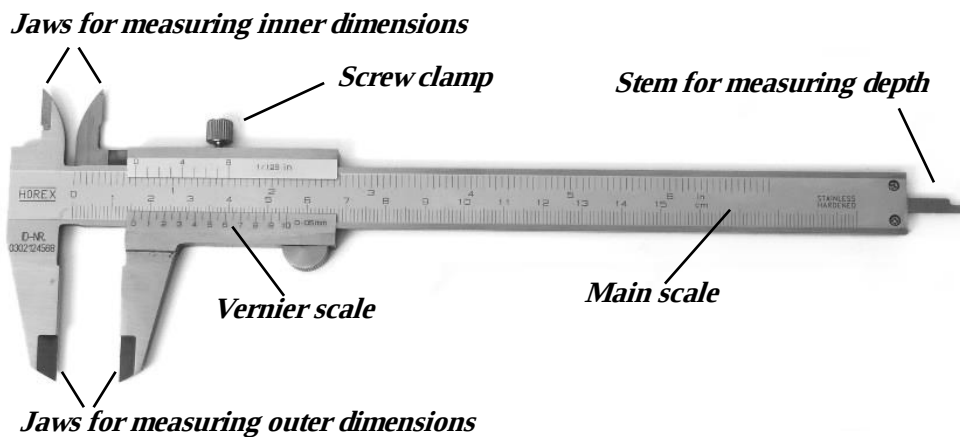


Figure 1.14: Main parts of a vernier caliper.

The vernier scale divides every millimeter into 10 divisions of $1/10 \text{ mm}$. Figure 1.15 gives an example of how to read a measurement from a vernier caliper.

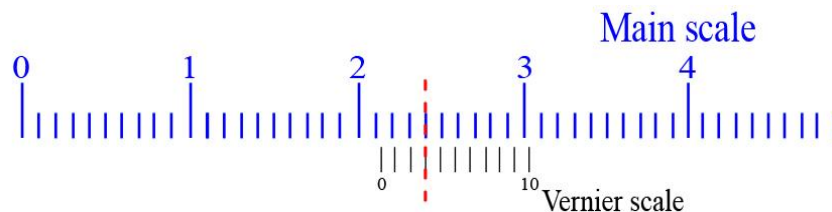


Figure 1.15: Reading example.

It is clear that the 0 mark on the vernier scale is a little bit shifted to the right from the 21 mm reading on the main scale. This means that the reading is expected to be between 21 and 22 mm. It can be seen that the fourth division of the vernier scale is aligned with the fourth division of the main scale. This means that the measured value exceeds the 21 mm by 0.3 mm. So that the final and exact reading is 21.3 mm.



The main and vernier scales each has mm and inch scale. Of course, the mm scale on the main scale should be used with the mm scale on the Vernier scale and the same for the inch scales. Do not mix up scales.

The Zero error :

After using the vernier caliper for a while, error in measurements might occur. This error is systematic and has to be determined.

When the caliper jaws are tightly closed **with nothing between them:**

- if the zero marks on both the main scale and the vernier scale both coincide as shown in Figure 1.16, this means there will be no zero error in the measurements.

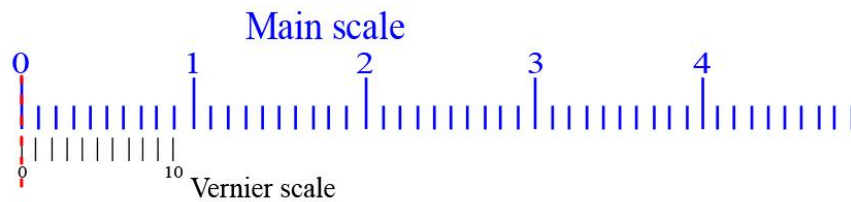


Figure 1.16: Vernier caliper with no zero error.

- If the zero marking of the vernier scale appears to the right of the zero marking of the main scale, this means there will be positive zero error in the measurements. The error is $x_{error} = +y$ cm, where y is the reading when the caliper jaws are tightly closed. To obtain the correct measurement, subtract this error from the measurement. The correct measurement in this case is

$$x_{correct} = x_{measured} - (x_{error}).$$

Figure 1.17 shows a vernier caliper with positive zero error $x_{error} = +0.2mm = +0.02cm$ when the caliper jaws are tightly closed.

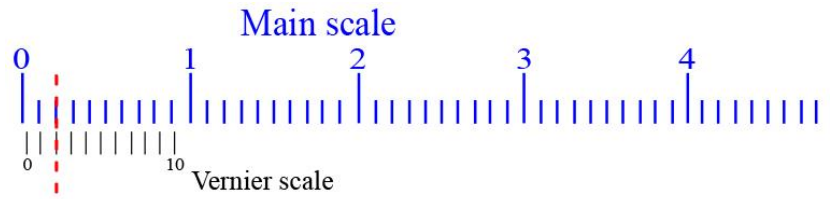


Figure 1.17: Vernier caliper with positive zero error.

- If the zero marking of the vernier scale appears to the left of the zero marking of the main scale, this means there will be negative zero error in the measurements. The error is $x_{error} = -(0.1 - y)$ cm, where y is the reading when the caliper jaws are tightly closed. To obtain the correct measurement, subtract this error from the measurement. The correct measurement in this case is

$$x_{correct} = x_{measured} + (x_{error}).$$

Figure 1.18 shows a vernier caliper with negative zero error, the reading $y = 0.8\text{mm} = 0.08\text{ cm}$. The correct measurement in this case is $x_{error} = -(0.1 - 0.08) = -0.02\text{cm}$ when the caliper jaws are tightly closed.

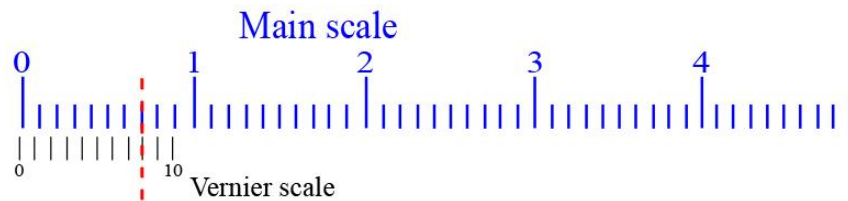


Figure 1.18: Vernier caliper with negative zero error.

2- Micrometer

A micrometer as the one shown in Figure 1.19 is one of the most accurate mechanical devices in common use.

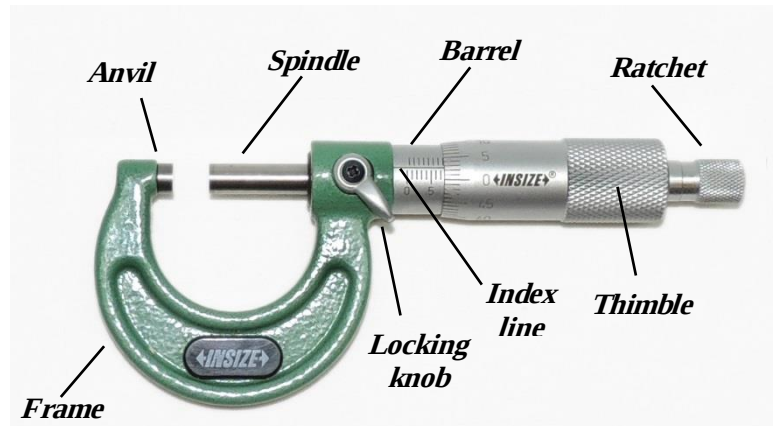


Figure 1.19: Main parts of a micrometer.

To take a measurement using a micrometer, the object to be measured should be placed between the anvil and the ratchet. Turning the ratchet will also turn the thimble and cause the spindle to move towards the anvil. The ratchet is turned until the measured object is enclosed between the spindle and the ratchet. Keep rotating the ratchet until a couple of clicks are heard, this means your reading is ready. You may use the locking knob to prevent the thimble from further rotations.

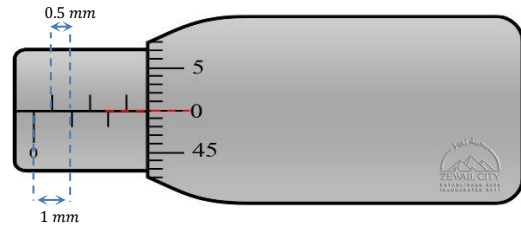


*On rotating the micrometer, always **use the ratchet NOT the thimble until a couple of clicks are heard**, otherwise, the measured object might get deformed or the micrometer itself might be damaged (zero error might appear in later measurements). In both cases, using the thimble to screw the micrometer will yield to incorrect measurements.*

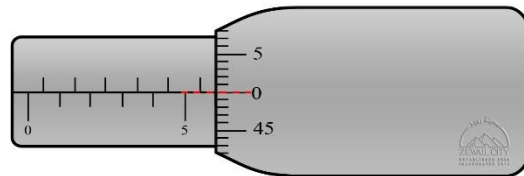
The barrel has horizontal markings, each representing 0.5 mm . The thimble has vertical markings all around it which divide every half a millimeter on the barrel into 50 divisions each of 0.01 mm . The index line on the barrel is used as a pointer for the scale on the

thimble. On turning the ratchet, you will notice that every 0.5 mm of the barrel scale, the index line coincides with the thimble 0 marking.

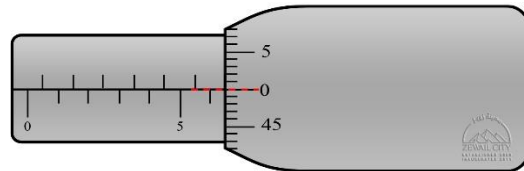
1. When the zero mark on the thimble coincides with the index line on the barrel, as shown, then the distance between the measuring rods is exactly an integer or half an integer number of millimeters. In the case shown, the distance between the anvil and spindle is **3.000 mm** . The line pointing downward on the barrel corresponding to 3 mm is barely visible under the thimble.



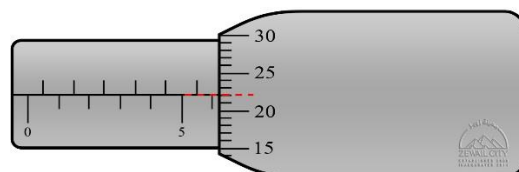
2. If the thimble is then rotated a couple of complete revolutions so it moves to the right to be as shown, then the distance between the anvil and spindle will be **6.000 mm** .



3. One more complete revolution takes the measured distance to **6.500 mm** , as shown. The line pointing upward on the barrel corresponding to 6.5 mm is now barely visible.

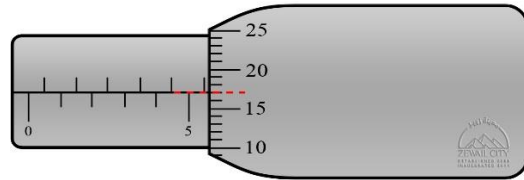


4. If we now turn the thimble just a little bit more so it moves to the left, we end up with the measurement shown. The distance is now 0.220 mm (as read from the barrel scale) above 6.000 mm , so the distance between the between



the anvil and spindle is now
6.220 mm.

5. If we rotate the thimble some more, we end up with the micrometer in an adjustment as shown. The distance is 0.170 mm above 5.500 mm. The measured distance is now $5.000 + 0.500 + 0.170 = \mathbf{5.670 \text{ mm}}$.



The Zero error :

After using the micrometer for a while, error in measurements might occur. This error is systematic and has to be determined. The procedure has some similarities to that used to determine the zero error for the vernier.

When the measuring rods are tightly closed **with nothing between them:**

- if the index line on the barrel coincides with the zero marking on the thimble as shown in Figure 1.20, this means there will be no zero error in the measurements.



Figure 1.20: Micrometer with no zero error.

- If the thimble edge is to the right of the zero marking of the barrel (i.e. the zero marking of the thimble scale is below the index line on the barrel), this means there will be positive zero error in the measurements. The error is $x_{error} = +y \text{ mm}$, where y is the reading when the measuring rods are tightly closed. To obtain the correct measurement, subtract this error from the measurement. The correct measurement in this case is

$$x_{\text{correct}} = x_{\text{measured}} - (x_{\text{error}}).$$

Figure 1.21 shows a micrometer with positive error $x_{\text{error}} = +0.03\text{mm}$ when the measuring rods are tightly closed.

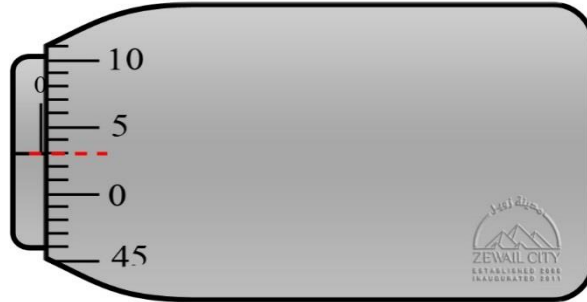


Figure 1.21: Micrometer with positive zero error.

- If the thimble edge is to the left of the zero marking of the barrel (i.e. the zero marking of the thimble scale is above the index line on the barrel), this means there will be negative zero error in the measurements. The error is $x_{\text{error}} = -(0.50 - y)$ mm, where y is the reading when the measuring rods are tightly closed. To obtain the correct measurement, subtract this error from the measurement. The correct measurement in this case is

$$x_{\text{correct}} = x_{\text{measured}} - (x_{\text{error}}).$$

Figure 1.22 shows a micrometer with negative zero error, the reading $y = 0.48\text{mm}$. The correct measurement in this case is $x_{\text{error}} = -(0.50 - 0.48) = -0.02\text{mm}$ when the measuring rods are tightly closed.

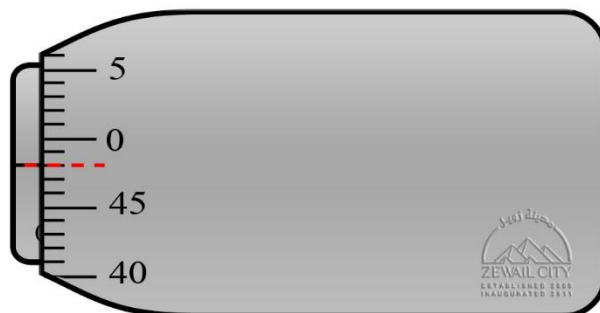


Figure 1.22: Micrometer with negative zero error.

3- Digital balance

A digital balance as shown in Figure 1.23 is used to measure the mass of any object with certain given resolution. Since it has a digital readout, we can take advantage of it to determine the zero error electronically.



Figure 1.23: Digital balance.

- 1- To start the measurement, press the “Zero\Power” button, the digital scale lights up.
- 2- Before performing the measurement, the scale should be set to zero. If the scale is found to be reading any value (either positive or negative), press the “Zero\Power” to remove the zero error. The balance positive and negative zero errors are shown in Figure 1.24 (a) and (b) respectively, while Figure 1.24 (c) shows the digital balance after removing the zero error

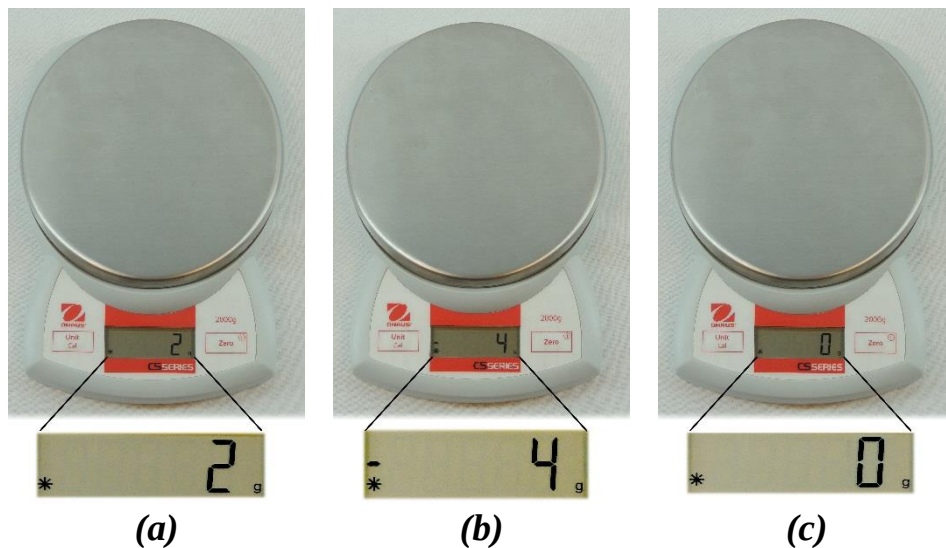


Figure 1.24: Digital balance before and after removing zero errors.

- 3- Every scale has a defined resolution for its measurements, the resolution of the balance can be found on label at the bottom of the balance.



It's a good practice – and a must - to always check the resolution of any measuring device you work with to expect the error in your measurements.

- 4- The digital balance can measure the mass/weight with various units. To obtain correct and reasonable readings, always make sure the balance is measuring the mass/weight with the desired unit. The “Unit” button switches between various units such as gm, Newtons, oz and lb. The adjusted unit is always displayed on the screen as shown in Figure 1.25.

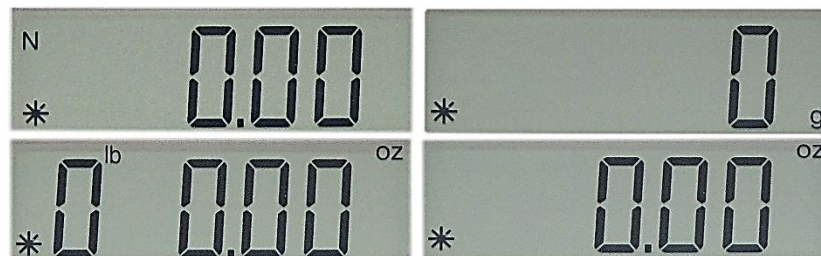


Figure 1.25: Various units measured by the digital balance.

- 5- After placing any mass on top of the digital scale, the reading on the screen will fluctuate for a few seconds until the scale shows an asterisk on the lower left corner of the screen when the reading is final as shown in Figure 1.26.



Figure 1.26: An asterisk on the lower left corner of the screen appears when the reading is final.



If you once use a digital scale that has no zero reading set button, you will have to put the zero error into

Appendix 2

For a cylinder of radius r and height h , the density is given by the equation:

$$\rho = \frac{m}{V} \quad (1)$$

Where m is the mass of the cylinder, and V is its volume. But,

$$V = \pi r^2 h = \frac{\pi D^2 h}{4} \quad (2)$$

Where $D=2r$ is the diameter of the cylinder.

from the above two equations we get:

$$\rho = \frac{4}{\pi} \frac{m}{D^2 h} \quad (3)$$

To get the error $\Delta\rho$ in the density ρ due to the error in measuring the quantities m, D and h , we can use the derivative equation,

$$d\rho = \left(\frac{\partial\rho}{\partial m}\right) dm + \left(\frac{\partial\rho}{\partial D}\right) dD + \left(\frac{\partial\rho}{\partial h}\right) dh \quad (4)$$

Where $\frac{\partial\rho}{\partial m}$, $\frac{\partial\rho}{\partial D}$ and $\frac{\partial\rho}{\partial h}$ is the partial derivative of density according to mass, diameter and height respectively (the partial derivative supposes all parameters are constants except the one you derivative with), then,

$$\frac{\partial\rho}{\partial m} = \frac{\partial}{\partial m} \left(\frac{4}{\pi} \frac{m}{D^2 h} \right) = \frac{4}{\pi} \frac{1}{D^2 h} \quad (5a)$$

$$\frac{\partial\rho}{\partial D} = \frac{\partial}{\partial D} \left(\frac{4}{\pi} \frac{m}{D^2 h} \right) = \frac{4}{\pi} \frac{m}{h} * (-2D^{-3}) \quad (5b)$$

$$\frac{\partial\rho}{\partial h} = \frac{\partial}{\partial h} \left(\frac{4}{\pi} \frac{m}{D^2 h} \right) = \frac{4}{\pi} \frac{m}{D^2} * (-h^{-2}) \quad (5c)$$

And equation (4) can be written as,

$$d\rho = \left(\frac{4}{\pi} \frac{1}{D^2 h} \right) \sigma_m + \left(\frac{4}{\pi} \frac{m}{h} * (-2D^{-3}) \right) \sigma_D + \left(\frac{4}{\pi} \frac{m}{D^2} * (-h^{-2}) \right) \sigma_h$$

divided this equation by to ρ , we get:

$$\frac{d\rho}{\rho} = \frac{\left(\frac{4}{\pi} \frac{1}{D^2 h} \right)}{\rho} \sigma_m + \frac{\left(\frac{4}{\pi} \frac{m}{h} * (-2D^{-3}) \right)}{\rho} \sigma_D + \frac{\left(\frac{4}{\pi} \frac{m}{D^2} * (-h^{-2}) \right)}{\rho} \sigma_h$$

as $\rho = \frac{4}{\pi} \frac{m}{D^2 h}$, so:

$$\frac{d\rho}{\rho} = \left(\frac{dm}{m} \right) - 2 \left(\frac{dD}{D} \right) \sigma_D - \left(\frac{dh}{h} \right) \quad (6)$$

The last equation can be written as,

$$\frac{\Delta \rho}{\rho} = \left(\frac{\Delta m}{m} \right) - 2 \left(\frac{\Delta D}{D} \right) \sigma_D - \left(\frac{\Delta h}{h} \right) \quad (7)$$

To ensure that the percentage error is equal to or more than the true error, we take the absolute values of the terms of the last equation ,i.e,

$$\left| \frac{\Delta \rho}{\rho} \right| = \left| \frac{\Delta m}{m} \right| + 2 \left| \frac{\Delta D}{D} \right| \sigma_D + \left| \frac{\Delta h}{h} \right| \quad (8)$$

Which gives the maximum possible percentage error in the density. In the experimental work, ρ , m , D and h are the average values of the measurements, while Δ denotes to the absolute value of the shift of the measurement from the average value,

$$x = \frac{\sum_i x_i}{N}$$

$$\Delta x = \frac{\sum_i |\Delta x_i|}{N}$$

$$\Delta x_i = |\bar{x} - x_i|$$

where N is the number of trials.

According to the definition of the uncertainty error in the Probability theory, the percentage error can calculated from the following formula,

$$\left| \frac{\Delta \rho}{\rho} \right| = \sqrt{\left(\frac{\sigma_m}{m} \right)^2 + \left(2 \frac{\sigma_D}{D} \right)^2 + \left(\frac{\sigma_h}{h} \right)^2} \quad (9)$$

Where σ denotes the standard deviation,

$$\sigma_m = \sqrt{\frac{\sum_i (\Delta m_i)^2}{N}}$$

With similar expressions for σ_h and σ_D .

Practically, we can calculate the percentage error using the equations (8) and (9) and take the largest value.